

Problem Set 2

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Problem 1. Strong Connectivity

(a) In a directed graph G equivalence relation R on the vertices of G as follows: $u R v$ is true IFF there exists a directed walk from u to v and a directed walk from v to u . (Note: A walk from u to v is expressed as $u \rightarrow v$) Now to show that the given relation is an equivalence relation:

- **Reflexive:** For any vertex u , there always exists walk of length from u to u in both directions, so $u R u$.
- **Symmetric :** Suppose $u R v$, There then there exists $(u \rightarrow v)$ and $(v \rightarrow u)$, Which are exactly two walks required for relation $u R v$.
- **Transitive:** If $a R b$ and $b R c$ then we know there exists walks $(a \rightarrow b)$, $(b \rightarrow a)$, $(b \rightarrow c)$, $(c \rightarrow b)$ and by concatenating those walk we get $(a \rightarrow b \rightarrow c)$ which implies $(a \rightarrow c)$ and similarly, by doing the same for the other two walks we get $(c \rightarrow a)$. Hence $b R c$.

(b) By comparing the definition of strongly connected components and R we observe that The equivalence classes of R are exactly the strongly connected components.

Problem 2. Bijections via Inverses

(a)

Proof. Let $a_1, a_2 \in A$ such that $f(a_1) = f(a_2)$, now applying g to both side, $g(f(a_1)) = g(f(a_2))$ since $\forall a, g(f(a)) = a$, $a_1 = a_2$. So f is surjective. $\square$

(b)

proof by contradiction. Assume f is not surjective. Hence, $\exists b \in B$ such that $\forall a \in A, f(a) \neq b$. But $f(g(b)) = b$ is true for all $b \in B$ which is a contradiction since $g(b)$ is an element of A that maps to b . $\square$

(c) From parts (a) and (b), we know f is a function, total, injective, and surjective. Hence f is a bijection.

Problem 3. Increasing Correspondence

(a) Let $(a, b, c) \in A$. Now by the definition $f(a, b, c) = (c - b, b - a, a)$ and because $a \leq b \leq c$ we observe that $c - b \geq 0$, $b - a \geq 0$ and $a \geq 0$, which implies that for every valid input pair there is at least one output. Therefore, the given function is a total function.

Now we show that every output of f lies in set B . As we know $\forall (a, b, c) \in A$, $f(a, b, c) = ((c - b, b - a, a))$, and given pair is part of set B IFF

$$(c - b) + 2(b - a) + 3a = a + b + c = 1000$$

which satisfies all necessary conditions and by this we can conclude that $\forall (a, b, c) \in A$, so $f((a, b, c)) \in B$.

(b) Total function $g : B \rightarrow A$, $g(x, y, z) := (z, y + z, x + y + z)$
 $\forall (x, y, z) \in B$, triples (z, y, x) is defined over natural number and since addition is deterministic operation $(z, y + z, x + y + z)$ will garentes exactly one output for all possible input. Also $x + y + z \geq y + z \geq z$ is true because $y, x \geq 0$. Sum: $z + (y + z) + (x + y + z) = x + 2y + 3z = 1000$. So $g(x, y, z) \in A$.

$$g \circ f = g(f(a, b, c)) = g(c - b, b - a, a) = (a, (b - a) + a, (c - b) + (b - a) + a) = (a, b, c).$$

$$f \circ g = f(g(x, y, z)) = f(z, y + z, x + y + z) = ((x + y + z) - (y + z), (y + z) - z, z) = (x, y, z).$$

(c) Since we proved that g is a two-sided inverse of f , and that is necessary and sufficient to prove that f is bijection

Problem 4. Counting Graphs

(a)

$$2^{\frac{n(n-1)}{2}}$$

(b)

$$n \times 4 \times 3 \times 3 = 36n$$

Problem 5. Counting

(a) There are total $21!$ ways to arrange 21 consonants. Now, since every vowel appears to the left of a consonant, we must choose 5 total consonants to make a gap in front and place a vowel. There are $\binom{21}{5}$ ways to make such a gap, and now there are $5!$ ways to put a vowel into those gaps.

Hence total way =

$$21! \times \binom{21}{5} \times 5!$$

(b) We first need to make a total of 5 [VCC] boxes and the remaining 11 consonants in [C] boxes:

$$\frac{\left(\binom{21}{2} \times \binom{19}{2} \times \binom{17}{2} \times \binom{15}{2} \times \binom{13}{2} \times 2^5\right)}{5!} \times 5! \times 16!$$

We needed to divide by $5!$ because the count is for distinct sections of groups, meaning (bc, df,...) is the same as (df, bc,...) and 2^5 because pairs bc and cb are distinct. and there are $5!$ ways to place vowels in each box. and there are a total of 11 boxes of [C], so the number of ways to arrange 16 distinct boxes is $16!$

And simplifying the given equation gives:

$$\frac{21! \times 16!}{11!}$$

(c) (Note: Another concise way to express the equation in (b) is used here, which is via the multinomial coefficient) Need to choose n partitions each of size 2, from $2n$ choices that can be done in:

$$\binom{2n}{2, 2, 2, \dots, 2} \times \frac{1}{n!} = \frac{(2n)!}{2^n n!}$$

We must here, just like in part (b), the binomial coefficient gives an ordered pair, and since there are a total of n terms, there are $n!$ copies of same arrangement.

(d) We can simply apply the Stars and Bars combinatorics theorem. Here we have n stars and 9 bars, (since there are 10 digits 0 to 9 and each partition says how many times a digit appears in the number) and we

need to find the number of ways to arrange them. The total number of positions to place bars is $n + 9$ and we need to choose 9 positions for bars, which gives us the following expression:

$$\binom{\text{total position to place bars}}{\text{bars}} = \binom{\text{stars+bars}}{\text{bars}} = \binom{n+9}{9}$$